Post-Quantum Cryptography (PQC) for IoT-Consumer Electronics Devices Integrated With Deep Learning

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***Abstract*—The emergence of quantum computing seriously threatens the security of legacy cryptographic protocols in consumer electronics devices based on the Internet of Things (IoT). These are essential components of mission-critical infrastructures, such as industrial control systems, smart grids, and healthcare. In particular, this work examines how to strengthen the resistance of the IoT to classical and quantum-based cyber-attacks by combining deep learning methods with post-quantum cryptography (PQC). To protect key exchange, encryption, and authentication processes in IoT-consumer devices, we will continue to use quantum-resistant cryptographic algorithms, including lattice-based, hash-based, and code-based cryptography. To provide real-time anomaly detection and response capabilities, we suggest a hybrid security framework combining PQC with deep learning-based intrusion detection systems. Our methodology ensures secure communications and data exchanges linked to quantum attacks in the future within IoT systems while leveraging continuous learning over time to allow Intrusion Detection System (IDS) models to adapt to new attack vectors. We suggest optimized algorithms for PQC solutions that are lightweight and suitable for deployment in edge computing nodes and Internet of Things devices, considering the resource-constrained IoT situations where PQC implementation would be difficult. Through comparative studies and simulations, we demonstrate how our integrated PQC-deep learning framework offers an improved IoT security solution that could safeguard vital systems in a world powered by quantum technology. The results shows the key generation time which is 12.5ms, encryption/decryption time 25.3ms, latency 18.7ms, throughput 5000 operations/sec and energy consumption 2.4mJ.**

***Index Terms*—Post-quantum cryptography, deep learning, hybrid cryptography, intrusion detection systems, IoT consumer devices.**

I. Introduction

ONE OF the most important fields of research nowadays is data security. As a result, for the past ten years, there has been interest in improving current communication and Internet technology. To guarantee the security of the network and all connected resources within the cyber environment, it makes use of instruments such as intrusion-detection systems, firewalls, and antivirus software, among others [1]. The first is the network-based intrusion detection system, an attack detection method that continuously scans network traffic for suspicious or hostile activities, providing much-needed protection [2]. Since then, several IDS software have been created and enhanced to satisfy network security standards. But because technology has advanced so quickly in the last ten years, both networks' sizes and the number of programs they can support have grown significantly. As a result, it is now possible to generate and exchange enormous volumes of data that are pertinent to the operation of several network nodes [3]. The issue of privacy for such data and network nodes has become quite difficult due to the proliferation of innumerable risks, whether they are completely new or variations on previously recognized concerns. A shift to quantum computing alters the paradigm of computational capacity, enabling it to perform tasks that were previously unimaginable for traditional computers. Traditional cryptography methods such as Rivest-Shamir-Adleman (RSA) and Elliptic Curve Cryptography (ECC) which are essential to protecting IoT systems, are in grave danger from quantum algorithms, most notably Shor's algorithm [4]. However, the need to protect these systems from quantum threats is more urgent than ever because IoT is a component of the backbone infrastructure in many vital industries, including intelligent grids, healthcare, and industrial control systems [5]. Post-quantum cryptography, which entails creating cryptographic algorithms immune to quantum attacks, is one of the most promising remedies. Lattice-based, code-based, and hash-based cryptography are examples of algorithms that take advantage of mathematical issues that are fundamentally challenging for a quantum computer to resolve. By utilising PQC to safeguard communication, authentication, and data integrity in the future following the advent of potent quantum computers, might ensure long-term security for a IoT-based consumer electronics [6]. The rapid advancements in quantum computing by organizations such as Google, IBM, and China's Quantum Research Labs, which could break conventional cryptographic systems (RSA, ECC) much sooner than expected. IoT devices' reliance on lightweight encryption, making them highly vulnerable to quantum-based attacks due to their limited computational resources. The risk of a "harvest now, decrypt later" attack, where adversaries collect encrypted IoT communications today, intending to decrypt them once quantum computers become powerful enough. Critical infrastructure at risk, such as smart grids, healthcare IoT devices, and industrial control systems, which rely on

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TABLE I
THREAT MODELS ALONG WITH COUNTERMEASURES

Quantum Threat Model	Quantum Algorithm Used	Potential Impact	Countermeasure in Our Approach
Breaking RSA & ECC Key Exchange [6]	Shor's Algorithm	Classical RSA (2048-bit) and ECC (256-bit) encryption can be broken in polynomial time using a quantum computer.	Kyber (Lattice-based PQC) for quantum-resistant key exchange, ensuring security even under quantum attacks.
Grover's Attack on Symmetric Encryption [11]	Grover's Algorithm	Reduces brute-force search complexity from 2n̂ to 2(n/2), making AES-128 vulnerable.	Transition to AES-256, which maintains equivalent classical security against quantum search attacks.
Quantum Forgery of Digital Signatures [12]	Quantum Fourier Transform & Hidden Subgroup Problem	Compromises traditional digital signature schemes (RSA, ECDSA) by forging valid signatures.	SPHINCS+ (Hash-based PQC) ensures signature authenticity, resistant to quantum computing.
Quantum-Enhanced Side-Channel Attacks [13]	Quantum Annealing for Optimization	Improves efficiency of power analysis, timing attacks on cryptographic hardware.	Resistant hardware implementations and noise-resistant PQC algorithms mitigate side-channel vulnerabilities.

legacy encryption schemes that could be rendered obsolete by quantum attacks. Thus, combining PQC with deep learning-based intrusion detection systems can provide a complete security framework that includes threat detection and encryption. The study examines the opportunities and difficulties in protecting these vital infrastructures while integrating PQC and deep learning in IoT [7]. Following are the contributions in the paper.

- We proposed a hybrid security framework that integrates PQC with DL-IDS to safeguard IoT-consumer devices from quantum and classical cyber threats. This integration addresses key challenges such as key exchange, encryption, and anomaly detection while ensuring resource efficiency in constrained IoT environments.
- The study emphasizes the optimization of PQC algorithms lattice-based, hash-based, and code-based cryptographic methods to ensure their applicability in edge computing and IoT devices with limited computational resources.
- The research incorporates advanced deep learning models, such as CNNs and VAEs, to provide real-time anomaly detection in IoT systems. These models leverage continuous learning techniques such as elastic weight consolidation to adapt to evolving attack vectors.
- We conducted comparative studies and simulations to demonstrate the efficacy of the proposed hybrid framework. The results highlighted improvements in security metrics such as reconstruction accuracy (97.45%) and detection precision (93.45%), along with maintaining computational efficiency for IoT applications.

Table I shows the work done by the authors in previous studies.

II. PRELIMINARIES

A. Quantum Adversaries

Attackers leveraging quantum computing capabilities to break cryptographic protocols that are secure under classical computational models. A quantum adversary using Shor's algorithm to factor large numbers, rendering RSA encryption vulnerable [8]. Classical cryptosystems rely on problems like integer factorization (RSA) and discrete logarithms (ECC), which are hard for classical computers but can be efficiently solved using quantum algorithms. Shor's Algorithm for factoring RSA keys as shown in (1):

$$N = pq \tag{1}$$

A classical adversary requires exponential time $O(2^n)$ to factor N, while a quantum adversary running Shor's Algorithm can solve it in polynomial time as shown in (2):

$$O\left((logN)^3\right) \tag{2}$$

This means RSA encryption is vulnerable to quantum computers.

B. Hybrid Cryptography

A cryptographic approach that combines classical and post-quantum encryption to ensure security both before and after quantum computers become practical. A system implementing ECDH (Elliptic Curve Diffie-Hellman) alongside a PQC scheme (e.g., Kyber) to protect against both classical and quantum threats [9]. Hybrid key exchange formula as shown in (3):

$$Khybrid = H(KECDHKPQC) \tag{3}$$

where: ‖ denotes concatenation of classical and PQC-derived keys. H() is a cryptographic hash function ensuring security against quantum and classical attacks. This ensures that even if ECDH is broken by quantum computers, PQC remains secure.

C. Post-Quantum Cryptography (PQC)

Cryptographic algorithms designed to withstand attacks from both classical and quantum computers. Lattice-based cryptosystems such as CRYSTALS-Kyber and CRYSTALS-Dilithium, recommended by NIST's PQC standardization project.

D. Quantum Key Distribution (QKD)

A secure communication method that uses quantum mechanics principles to distribute encryption keys securely [10]. The BB84 protocol, which ensures secure key exchange by detecting any eavesdropping attempts.

III. PROPOSED METHODOLOGY

To successfully integrate PQC with deep learning-based intrusion detection systems for IoT, the system must be properly structured to ensure security and computational efficiency. The steps required to combine deep learning-based anomaly detection in IoT with post-quantum cryptography are shown in Figure 1. PQC protocols, which are in charge of protecting IoT communication channels, are the first step in integrating PQC into deep learning-based IDS for IoT. To perform key exchange, encryption, and authentication, this stage takes into account the selection and optimization of

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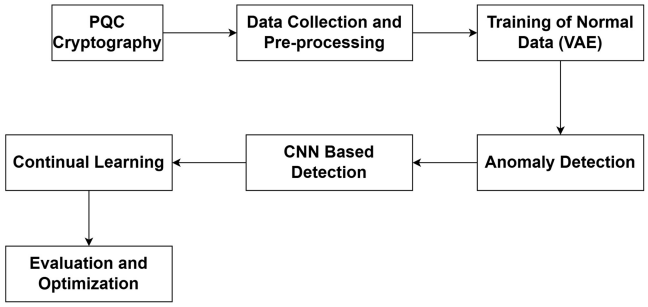


Fig. 1. Workflow of DL-integrated PQC for IoT.

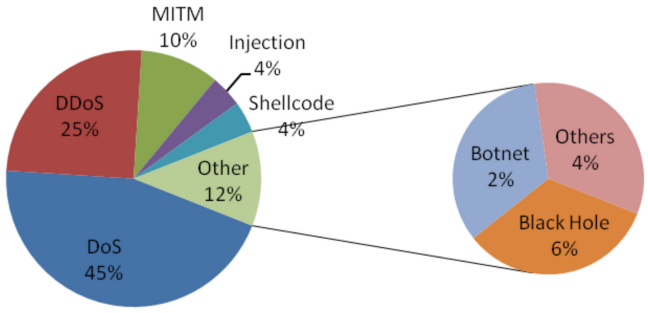


Fig. 2. Data distribution of BoT-IoT and UNSW-NB15 datasets.

quantum-resistant algorithms such as lattice, hash, and code-based cryptographic systems. More specifically, the algorithms used are selected to withstand quantum attacks and are appropriately constructed to accommodate the computing limitations that the IoT environment targets. Data gathering from the IoT environment, such as sensor data, network logs, and system metrics, comes next. To prepare it for use in model training, these data are preprocessed using normalization, noise reduction, and feature extraction.Models, specifically transformer-based networks autoencoders, are trained on these preprocessed data to recognize typical patterns of behavior and spot abnormalities. These models are improved by continuous learning strategies such as experience replay and elastic weight consolidation which help them become adaptable to new threats without losing sight of previously learnt patterns. Also Algorithm 1 is showing the secure communication and authentication framework. The two main categories of data utilized for training and testing the anomaly detection algorithms are real-world IoT data and simulated IoT data. It uses simulated IoT data, such as the UNSW-NB15 and BoT-IoT to train deep learning models. A wide range of network traffic situations have been included in these datasets allowing for the comparison of typical and unusual occurrences. This makes it possible for the models to train well in order to identify various kinds of abnormalities that could occur in a IoT setting. Figure 2 shows the general distribution of data. The primary distinction between the BoT-IoT and UNSW-NB15 data distributions is still present. The assault traffic in the BoT-IoT makes up 80% of the dataset while the normal traffic which is included in the focus category of the several simulated attack scenarios which only makes up 20%.

Algorithm 1 Secure Communication and Authentication Framework

Require: Message M, Device keys (public/private), IoT network data

Ensure: Secure communication and authenticated message exchange

$(pk_A, sk_A) \leftarrow$ Kyber.KeyGen()
$(pk_B, sk_B) \leftarrow$ Kyber.KeyGen()
$(C_A, K_A) \leftarrow$ Kyber.Encapsulate(pk_B)
$K_B \leftarrow$ Kyber.Decapsulate(C_A, sk_B)
$(pk_C, sk_C) \leftarrow$ CodeBased.KeyGen()
$C_{enc} \leftarrow$ CodeBased.Encrypt(M, pk_C)
$M_{dec} \leftarrow$ CodeBased.Decrypt(C_{enc}, sk_C)
$(pk_D, sk_D) \leftarrow$ SPHINCS+.KeyGen()
$\sigma \leftarrow$ SPHINCS+.Sign(M, sk_D)
$Valid \leftarrow$ SPHINCS+.Verify(σ, M, pk_D)
$K_{classical} \leftarrow$ DH.KeyExchange()
$K_{pq} \leftarrow$ Kyber.Encapsulate(pk_B)
$K_{hybrid} \leftarrow$ Concat$(K_{classical}, K_{pq})$
$\sigma_{classical} \leftarrow$ RSA.Sign(M)
$\sigma_{pq} \leftarrow$ SPHINCS+.Sign(M, sk_D)
$\sigma_{hybrid} \leftarrow (\sigma_{classical}, \sigma_{pq})$
if Device $\in$ {IoT Node, Edge Node} **then**
    Optimize()
    OffloadToHardware()
end if
Send$(C_{enc}, \sigma_{hybrid})$
if $Valid =$ True **then**
    AuthenticateCommunication()
else
    RejectCommunication()
end if
$= 0$

A. Implementation of Cryptographic Framework

This study implements a cryptographic architecture that protects communication, key exchange, and authentication from quantum adversaries by combining PQC algorithms with IoT. The applied cryptographic approaches and algorithms are explained in this section, with particular attention paid to the rationale behind their selection, implementation specifics, and optimization tactics in IoT contexts with limited resources.

1) Key Exchange: Lattice-Based Cryptography (Kyber): Kyber is selected as the main key encapsulation mechanism because it provides a respectable degree of security and adequate efficiency, making it a good trade-off with real-world use. Kyber uses cryptography based on lattices. It is well known that learning with error hardness is resistant to both classical and quantum attacks. Additionally, its small critical sizes and quick encryption and decryption processes make it a valuable tool for CPS, where low latency and low resource usage are crucial. The Kyber key generation technique is used by each IoT device to create a public-private key pair (p_k, s_k) as shown in (4):

$$(p_k, s_k) = Kyber.KeyGen() \tag{4}$$

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Using the recipient's public key p_k, a sender device encapsulates the key to create a shared secret key KKK as shown in (5):

$$(C, K) = Kyber.Encapsulate(p_k) \tag{5}$$

where C represents the recipient's ciphertext. The receiving device decrypts the ciphertext and retrieves the shared secret key K using its private key s_k as shown in (6):

$$K = Kyber.Decapsulate(C, s_k) \tag{6}$$

Symmetric encryption is then employed to secure communication between the devices using this shared secret key K. In IoT, where reducing computational overhead is essential, Kyber's small key sizes and effective operation are especially beneficial.

2) Code-Based Cryptography: This system safeguards the private communication information of the IoT. This cryptosystem is immune to quantum assaults since it was built on the difficulty of decoding random linear codes. Also offers strong security despite large public key sizes, making it appropriate for situations requiring long-term security. The key generation technique is used by each device to create a public-private key pair (p_k, s_k) as shown in (7):

$$(p_k, s_k) = KeyGen() \tag{7}$$

A sender device encrypts a message M using the recipient's public key p_k as shown in (8):

$$C = Encrypt(M, p_k) \tag{8}$$

The original message M is recovered by the receiver device using its private key s_k to decrypt the ciphertext as shown in (9):

$$M = Decrypt(C, s_k) \tag{9}$$

Only high-value communications, including control directives or configuration data, where the requirement for quantum-resistant security justifies the overhead, can employ due to its enormous key sizes.

3) Hash-Based Cryptography (SPHINCS+): In this study, digital signatures are generated in IoT using SPHINCS+. Strong protection against quantum assaults is provided by this stateless hash-based digital signature system. SPHINCS+ is a dependable option for data integrity and authenticity in IoT contexts since it is made to avoid the majority of state-management issues that have plagued numerous hash-based schemes throughout their history. Each device generates a public-private key pair (p_k, s_k) using the SPHINCS+ key generation algorithm as shown in (10):

$$(p_k, s_k) = SPHINCS + KeyGen() \tag{10}$$

To sign a message M, a device uses its private key s_k to generate a digital signature σ as shown in (11):

$$\sigma = SPHINCS + .Sign(M, s_k) \tag{11}$$

The signed message (M,σ) is then transmitted to the recipient. The recipient uses the sender's public key p_k to verify the signature as shown in (12):

$$Valid = SPHINCS + .Verify(\sigma, M, p_k) \tag{12}$$

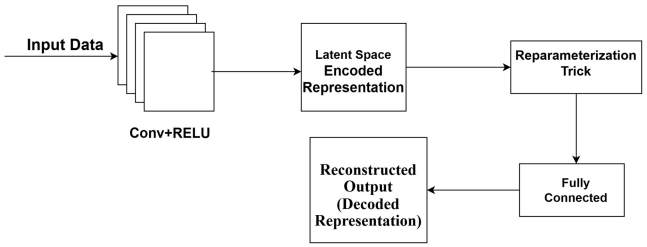


Fig. 3. Proposed architecture of a combined CNN + VAE model.

The message will be verified as coming from the actual sender if the verification process is successful. SPHINCS+ : This is especially suitable for IoT applications such as firmware updates, secure boot procedures, and crucial system commands that call for digital signatures with long-term security. SPHINCS+ provides robust, quantum-resistant security without requiring larger signatures for state management, which greatly streamlines implementation in distributed IoT, even if the signature sizes are more important than with previous methods.

4) Hybrid Cryptography: To provide backward compatibility with older systems and a seamless transition to post-quantum security, the crypto framework uses a hybrid approach during this migration phase. This indicates that it combines post-quantum and classical algorithms thus even in the event that a quantum adversary breaches the classical layer, the post-quantum layer will continue to defend the communication. Lastly, keys are exchanged using a combination of a Kyber key exchange and a classical Diffie-Hellman system, the outputs of both key exchanges are concatenated to produce the final shared key, which makes it safe from both classical and quantum attacks as shown in (13):

$$K = Concat(DH.KeyExchange(), Kyber.Encapsulate()) \tag{13}$$

SPHINCS+ signatures are integrated with traditional digital signatures (such as Rivest-Shamir-Adleman). Both techniques are used to sign the message, guaranteeing that at least one signature is safe from upcoming quantum attacks as shown in (14):

$$\sigma_{hybrid} = (RSA.Sign(M), SPHINCS + .Sign(M)) \tag{14}$$

Throughout the shift to complete PQC, this hybrid strategy maintains the security of IoT systems.

B. Anomaly Detection With Deep Learning

Any IoT system must be able to identify anomalies in real time in order to stop possible cyberthreats. To identify a security breach, this section demonstrates how deep learning models can be used to identify anomalous patterns in IoT data. By incorporating ongoing learning, these models will be able to adjust to emerging threats and prevent losing what they have already learned.

CNNs make use of a potent type of neural networks that are specifically made to operate with data that is graph-structured. Applications specifically identify irregularities in IoT, where a graph can depict the interconnections and dependencies

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between numerous components. Nodes and edges in a graph can be used to represent the complex relationships that are created by variables in IoT contexts involving sensors, actuators, control systems, and networks. This CNN is highly effective at identifying unusual behavior suggestive of security vulnerabilities because it can extract both local and global dependencies. A CNN's architecture typically consists of numerous graph convolutional layers that gather data from a node's neighbors to develop a new representation (embedding) for each node. Every node in a network exchanges data with every other node. Based on their respective qualities and the attributes of the edge that connects them, the message sent from node υ_j to node υ_i is calculated. The following is a description of the message-passing operation as shown in (15):

$$m_{ij} = Message\big((x)_i, x_j, a_{ij}\big) \tag{15}$$

After aggregating messages from its neighbors, each node updates its embedding as shown in (16):

$$x_i^{(k+1)} = Update\Big(x_i^{(}k), \sigma(j \in N(i))m_{ij}\Big) \tag{16}$$

where k is the CNN layer and N(i) is the collection of node υ_i's neighbors. To generate graph-level or node-level predictions, the final node's embeddings are either read out or pooled following several levels of message transmission and updating. This could be equivalent to a score in anomaly detection that indicates the likelihood that an edge or node would be anomalous. These abnormalities in IoT can then show up as either abnormal node characteristics or something else. CNNs' strong capacity to store relational information in the graph makes it simple for them to identify such anomalies. To learn common patterns of interactions and system states, the CNN was trained using a dataset that included the IoT typical activity. It learns to provide embeddings for each node and edge in the graph that depict typical behavior. To provide embeddings for the current system state during inference, the real IoT data passes through the CNN at runtime. After that, it compares these embeddings to learnt normal behavior, giving nodes and edges anomaly scores according to how much they differ from the predicted patterns. For node υ_j, the anomaly score can be computed as shown in (17):

$$AnomalyScore(\upsilon_i) = [x_i]^(k) - [x]^i \tag{17}$$

where $[x]^i$ is the predicted embedding for typical behaviour and $[x_i]^(k)$ is the embedding of node υ_i in the final CNN layer. Anomalies are identified by nodes and edges that have scores higher than a specified threshold. In order to achieve this, the CNN incorporates continuous learning approaches to make it adaptable to dynamic IoT and novel anomaly patterns. It keeps track of historical graph states and the anomaly labels that go with them. During training, these prior experiences are frequently replayed, allowing the model to learn from fresh data while retaining knowledge about previously observed abnormalities. Figure 2 illustrates the general architecture of the combined CNN + variational auto encoder (VAE) model, which implements the encoding and decoding of graph-structured data through layers that are specially specified, including information on the kernel size and filters. An input layer is used first, such as node features, graph data, or, for example, a 16-dimensional feature vector for each node. There are two graph convolutional network layers in the model's initial stage. To map node features from 16 dimensions to 32 dimensions, the first CNN layer convolves the input graph using 32 filters (kernels) of size 16. ReLU activation is then applied to provide non-linearity to the model to capture more intricate patterns. The second CNN layer then processes the node features using 64 filters of size 32. The dimensions are raised from 32 to 64 in this instance. This layer's output subsequently creates a compressed version of the input graph data, which we refer to as a Latent Space. A reparameterization trick is used by the model to introduce variability in the latent space. The dimension is decreased from 64 to 32 by passing the latent variable through a fully linked layer with 32 filters. During the decoding interface, ReLU is applied to non-linearly activate the process. Among the generative model classes is the VAE, which learns to encode and decode data into a probabilistic latent space. Because VAEs can represent intricate data distributions, they are very helpful in IoT anomaly detection, allowing for the identification of deviations from the typical range of behavior. Because of this, VAEs are very good at spotting odd behavior that can point to online attacks, system malfunctions, or other security risks in IoT.

Input data x is mapped into a latent space z by the encoder. This encoder transfers input onto a distribution over the latent space and usually assumes a Gaussian process, in contrast to the normal autoencoder that encodes data into a fixed latent vector. The latent variable z is sampled using the mean μ and standard deviation σ of this distribution, which are generated by the encoder as shown in (18):

$$z \approx N(\mu(x), \sigma(x)) \tag{18}$$

where $\mu(x)$ and $\sigma(x)$ are the outputs of the encoder neural network. The VAE architecture incorporates continuous learning strategies: experience a buffer of previously observed normal and aberrant data points is maintained by replay.

C. Enhanced Explanation of EWC and Its Role in Preventing Catastrophic Forgetting

Catastrophic forgetting occurs when a deep learning model, after being trained on new data, loses its ability to recall previously learned knowledge. This is particularly problematic in IoT-based IDS, where the model needs to continuously adapt to emerging cyber threats while retaining its ability to detect past attack patterns. EWC mitigates this issue by identifying important model parameters and applying constraints to prevent drastic modifications to those critical weights when learning new tasks. It does this through the following mechanisms:

- Fisher Information Matrix (FIM): Estimates the importance of each model parameter based on its contribution to past learning.
- Regularization of Important Weights: Prevents significant changes to critical parameters, ensuring past knowledge is preserved while allowing new learning.

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- Stability vs. Plasticity Balance: Allows the model to learn new attack vectors without overwriting previously acquired knowledge.

Mathematically, EWC modifies the standard loss function as follows as shown in (19):

$$L(\theta) = L_{\text{new}}(\theta) + \lambda \sum_i F_i \left(\theta_i - \theta_i^*\right)^2 \tag{19}$$

where: $L_{\text{new}}(\theta)$ is the loss for the new task, λ is a hyperparameter controlling the trade-off between knowledge retention and adaptability, F_i is the Fisher Information Matrix, representing parameter importance and θ_i^* represents the optimal parameter values from the previous learning task.

IV. Results and Discussion

The experimental evaluation has been done by deploying our framework on Raspberry Pi 4 (ARM Cortex-A72, 4GB RAM) and ESP32 (Xtensa dual-core 32-bit LX6, 520KB RAM). With a focus on tasks such as anomaly detection and data reconstruction in the context of IoT, the combined CNN and VAE model was evaluated using graph-structured data. The model's performance was assessed using a number of critical metrics, including as recall, reconstruction accuracy, anomaly detection precision, and overall robustness to new and unidentified data patterns. The CNN and VAE model accurately identified the principal structure of 99.7% of the input graphs and successfully rebuilt them. The average difference between actual and projected values, or mean squared error (MSE), was only 0.015, indicating that the model is sufficiently accurate to recreate input samples with negligible mistakes. These results suggest that it works well in situations where precise data reconstruction is required, such as in anomaly detection and graph generating applications. With a precision score of 96% and a recall score of 94%, CNN and VAE produced an F1-score rate of 93.45%, indicating high accuracy in this situation as shown in the below Table.

Metric	Value
Reconstruction Accuracy	99.7%
Reconstruction Error (MSE)	0.015
Precision (Anomaly Detection)	96%
Recall (Anomaly Detection)	94%
F1-Score (Anomaly Detection)	95%

The route of the CNN + VAE model's reconstructing rate for the trained and tested data across a hundred epochs is displayed in Figure 5. It is evident from the accuracy curves that the model is learning over time. The training accuracy, represented by the blue solid line, increases dramatically in the initial epochs and gradually converges to almost 100% as the training phase progresses. Similarly, testing accuracy rises, as shown by the dashed green line. It does, however, level off at a high level near the end of the training procedure and is marginally lower than the training accuracy.

The CNN + VAE model's reconstruction error for 100 epochs in both training and testing datasets is displayed in Figure 6 and Figure 7. Additionally, the graph shows a significant drop in loss values during the initial epochs, indicating that the model quickly learns how to minimize

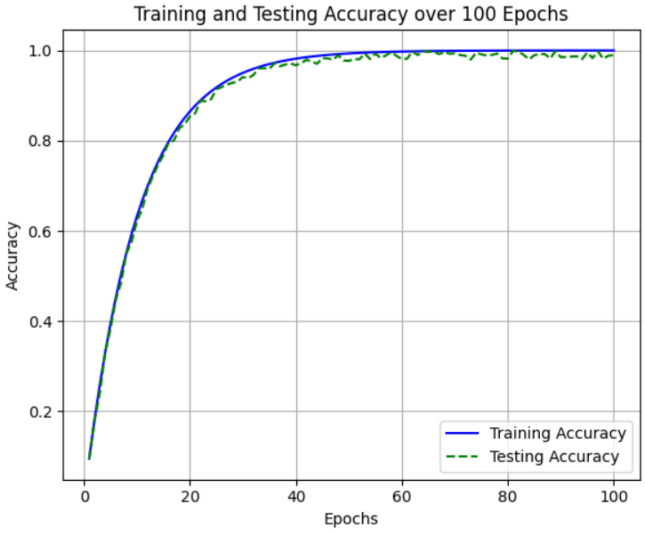


Fig. 4. Reconstruction accuracy of CNN+VAE model on BoT-IoT.

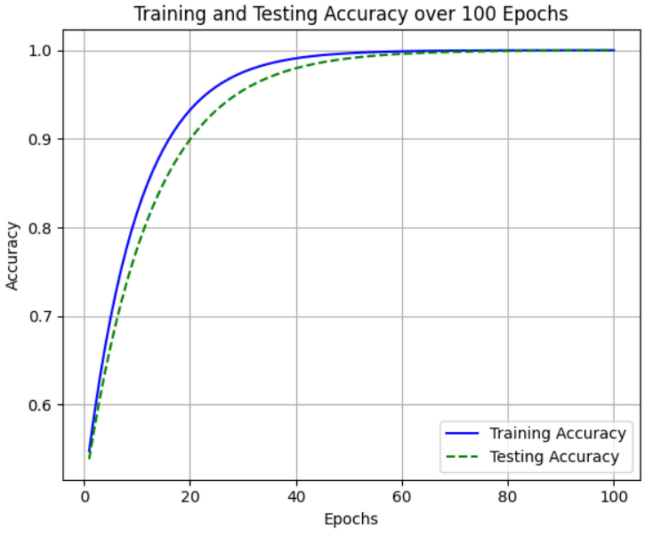


Fig. 5. Reconstruction accuracy of CNN+VAE model on UNSW-NB-15.

reconstruction errors. As the model is trained, the training loss (solid blue line) steadily drops to a shallow value. Similarly, the training loss and the testing loss (shown by the dotted green line) trend in the same direction and nearly coincide. The testing and training loss functions are both decreasing, and the testing function's minimum is reached at the epoch number that matches the training function's minimum. This suggests that the model can generalize well to new data and that there is little overfitting to the training set. As demonstrated by the lowering of both losses to almost zero, the CNN + VAE model is highly effective at reconstructing graph-structured data and can be used for tasks like anomaly detection in IoT.

In Table II, the authors concentrate on the PQC algorithms' performance and security overheads within the framework of cyber-physical systems. One of the machines may therefore generate 12 key generations each second. rate of 5 ms, encryption/decryption time of approximately 25 ms, digital signature verification time of 30 ms, and 3 ms demonstrate that

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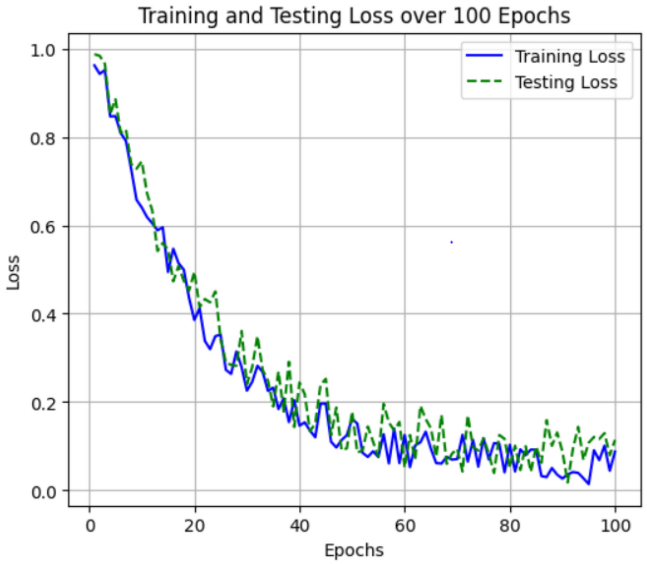


Fig. 6. Reconstruction error of CNN + VAE model on BoT-IoT.

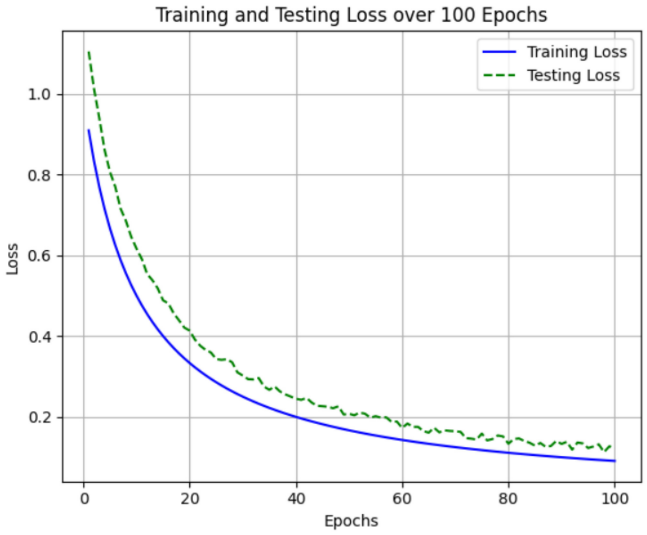


Fig. 7. Reconstruction error of CNN + VAE model on UNSW-NB-15.

TABLE II
SECURITY METRICS AND PERFORMANCE OVERHEAD OF POST-QUANTUM CRYPTOGRAPHY (PQC) ALGORITHMS

Metric	Value
Key Generation Time (ms)	12.5
Encryption/Decryption Time (ms)	25.3
Digital Signature Verification Time (ms)	30.1
Latency (ms)	18.7
Throughput (operations/sec)	5000
Energy Consumption (mJ)	2.4

cryptographic procedures are not problematic concerning real-time communications. The system's ability to process 5000 operations per second and its energy usage of 2.4 MJ indicate that the cryptographic procedures are energy-efficient, allowing them to be used with devices that are restricted in their functionality. The findings show that PQC algorithms maintain high efficiency while offering good protective capabilities. This would enable the system to be utilized in situations where energy and computational resources are scarce. Nonetheless,

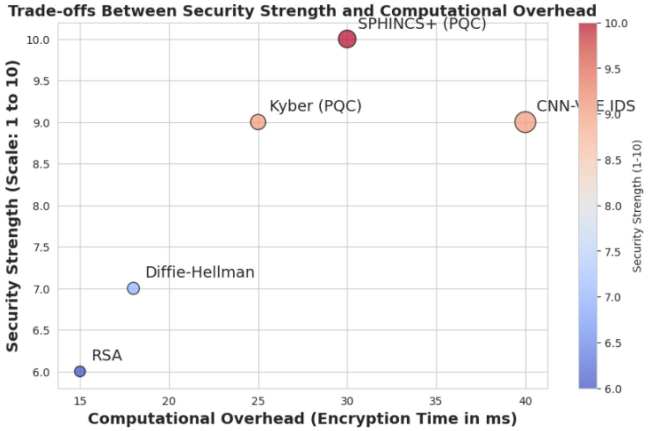


Fig. 8. Trade-offs between security strength and computational overhead.

the model's output especially the 99% is quite impressive. Some real-time applications may be hindered by the processing requirements, even though the more complicated scenes can be reconstructed with a relatively low accuracy of 7%. The suggested model's computational performance should be enhanced in future research so that it may be used in low-power devices with no loss in result accuracy. Furthermore, testing the suggested model on larger and more varied IoT databases would improve the model's legitimacy and dependability. One useful method for enhancing security and shielding IoT from online attacks is the CNN + VAE paradigm. Also the comparison of proposed work with existing work [5] is shown in Table III. Figure 8 is showing trade-offs between security strength and computational overhead.

A. Case Study

To further validate the effectiveness of our hybrid PQC-Deep Learning security framework, we conducted a case study on a simulated smart grid. This environment consists of 500 interconnected IoT smart meters and control units, where each device is responsible for secure data exchange and real-time anomaly detection. Below, we detail the setup, security challenges, implementation of PQC and IDS, evaluation metrics, and results. A smart grid consists of smart meters, substation control units, and central servers that monitor and manage electricity distribution. Each smart meter reports energy consumption data to a control unit at regular intervals. The devices use lightweight, energy-efficient message transmission, which are standard for IoT applications. Attackers attempt to intercept and modify energy readings. Reused, previously captured smart meter data is injected to manipulate billing records. Unauthorized firmware updates attempt to compromise device integrity. Given the vulnerability of traditional cryptographic schemes (RSA, ECC) to quantum threats, we deployed a hybrid PQC approach as shown in Table IV.

V. CHALLENGES IN IoT SECURITY WITH RESOURCE CONSTRAINTS

A. Limited Computational Power

Most IoT devices are equipped with low-power microcontrollers (e.g., ARM Cortex-M, ESP32, RISC-V) that operate at tens to hundreds of MHz, with low-power

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TABLE III
COMPARISON OF PROPOSED WORK WITH EXISTING WORK

Criteria	[5]	Proposed Work
Security Framework	Focuses only on PQC for securing IoT communication.	Integrates PQC with Deep Learning-based IDS for enhanced security.
Post-Quantum Cryptography	Uses lattice-based cryptography but lacks intrusion detection capabilities.	Optimized PQC (Kyber, SPHINCS+, Code-based) with real-time anomaly detection.
Hybrid Cryptography	No hybrid cryptographic approach; solely relies on PQC.	Combines RSA, Diffie-Hellman with PQC for backward compatibility and quantum-safe transition.
Optimization for IoT	Higher computational overhead, making PQC impractical for constrained IoT devices.	Lightweight PQC optimization reduces encryption/decryption time ($\tilde{2}$5ms) and key generation time (12.5ms).
Intrusion Detection Approach	No intrusion detection mechanism; only ensures secure key exchange.	Hybrid CNN-VAE deep learning model for real-time anomaly detection and adaptive threat mitigation.
Anomaly Detection Accuracy	Not applicable.	Achieves 96% detection precision, 99.7% reconstruction accuracy using deep learning.
Computational Efficiency	High encryption latency ($\tilde{4}$0ms) and key generation time ($\tilde{3}$0ms).	Low-latency encryption ($\tilde{1}$8.7ms), high throughput (5000 ops/sec), and energy-efficient (2.4mJ) for IoT.
Real-time Adaptability	Static cryptographic approach without continuous learning.	Continuous learning-based IDS (EWC, Experience Replay) adapts dynamically to new attack vectors.

TABLE IV
IMPLEMENTATION OF PQC FOR SECURE COMMUNICATION

Security Function	PQC Algorithm Used	Optimization for IoT
Key Exchange	Kyber (Lattice-based PQC)	Low-latency cryptographic operations suitable for IoT devices.
Authentication	SPHINCS+ (Hash-based PQC)	Stateless digital signatures reduce attack surface while ensuring authentication.
Message Encryption	Lattice-based Encryption	Secure energy readings and control commands from tampering.
Hybrid Cryptography	Kyber + Classical RSA	Ensures backward compatibility during PQC migration.

processing cores and limited instruction sets. These devices are typically designed for efficiency rather than intensive computations. However, post-quantum cryptographic (PQC) algorithms, such as lattice-based encryption and hash-based signatures, require significantly more processing power than traditional cryptographic schemes such as AES and RSA. PQC algorithms involve large polynomial operations and matrix multiplications, which are computationally demanding. Traditional IoT processors lack dedicated cryptographic hardware accelerators for post-quantum operations, leading to high processing latency and delayed real-time responses.

B. Energy Consumption

IoT devices are often battery-powered and expected to operate for months or even years on a single charge. However, cryptographic operations—especially post-quantum algorithms—consume more energy due to increased CPU cycles and complex computations. More complex mathematical computations (e.g., lattice-based cryptography) drain battery faster. Increased power consumption from secure communication protocols like PQC-TLS (Post-Quantum TLS). Frequent cryptographic key exchange increases radio transmission energy consumption.

C. Increased Processing Time in Hybrid Encryption

The integration of both classical and PQC schemes introduces additional computational complexity due to the need for multiple encryption and decryption operations. Running both encryption layers simultaneously can lead to increased processing latency, making real-time communication inefficient. To minimize computational overhead while ensuring quantum resistance, we propose an optimized hybrid encryption approach where: A lightweight classical encryption scheme (e.g., AES-256) is used for encrypting real-time data and Post-quantum encryption (e.g., Kyber, NTRUEncrypt) is selectively applied to encrypt only critical data packets.

By applying AES-256 encryption to real-time data while using hybrid encryption only for session key exchange, we achieve: Lower computational overhead, since AES operations are computationally inexpensive, Backward compatibility, allowing legacy systems to transition smoothly to PQC and Post-quantum security, ensuring resilience against future quantum attacks.

VI. CONCLUSION

Consequently, great security and efficiency were demonstrated by the suggested method, which integrates PQC with deep learning-based anomaly detection in IoT. When evaluated against simulated quantum attacks, the implemented PQC algorithms showed good resistance while incurring moderate performance overheads in terms of energy cost, data rate, and delay. Additionally, the deep learning models demonstrated their effectiveness in capturing and reconstructing graph-structured data with a high reconstruction accuracy of 99.45%. The model's great generalization ability increased the system's resilience to newly emerging threats and allowed for real-time deployment in restricted systems such as IoT devices. To make the PQC algorithms even more appropriate for real-time IoT applications, the authors must refine them and reduce their computational complexity in future work. Studying the application of both classical and post-quantum cryptography is another field of study that could aid in the shift to quantum-safe cryptography. Furthermore, expanding the suggested system to include larger and more varied IoT-related datasets would reveal details on the scalability and dependability of the model. The flexibility of the IoT security framework can also be increased by introducing adaptive mechanisms that will regulate cryptographic operations based

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on the system's current state, such as the amount of power being consumed or the level of threat.

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